

Josiah Holder

R&D Engineer: sensors, test systems, and embedded electronics

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SUMMARY

R&D engineer with 3.5 years developing agricultural sensors from concept through production. Builds the mechanical designs, test systems, and data analysis that show whether a design meets spec. Experience in high-speed data acquisition, CAN, and Python analysis tools, plus working directly with farmers, dealers, and suppliers.

SKILLS

Design and CAD: Creo, Fusion 360, DFM, GD&T drawings, tolerance stack-up, injection molded, SLS, and machined part design

Test and validation: Test stand design, DAQ, thermal chamber testing, CPK and PPK studies, statistical validation

Software and data: Python, Julia, C#, Git, time-synchronized data pipelines, AI-assisted development

Electronics: KiCad (4-layer PCB), SPI and I2S firmware, high-speed ADC sampling, CAN and CAN FD analysis, LTspice

EXPERIENCE

Precision Planting: R&D Engineer

Jan 2025 - Present, Fargo, ND

- Designed the validation system (test stand, control software, data pipeline, analysis) for an optical seed-flow sensor, proving the design met its $\pm 2\%$ requirement, with 95% of sensors within $\pm 2\%$ of the mean
- Cut sensor-to-sensor variance about 3x by tightening the design to reduce tolerance stack-up, and extended measurement range about 80%
- Ran 50+ sensors on the bench and analyzed data from 250+ installed on farmer equipment over two seasons
- Developed three mechanical embodiments (injection molded, SLS, machined), quoted them with multiple manufacturers, and chose the lowest-cost option with no loss in usability
- Designed a 4-layer KiCad board and firmware for a high-speed data acquisition prototype using an AI-assisted workflow; both revisions met their requirements
- Wrote CAN and CAN FD parsing and monitoring tools (ValueCAN, Kvaser) that produce time-synchronized test data
- Built a Python STEP assembly viewer, packaged as a standalone app, for product support, saving an estimated 4+ engineering hours per product
- Visit farms and dealers to gather feedback, and work with suppliers and contract manufacturers on cost and end-of-line testing

Appareo Systems (Intelligent Ag): R&D Engineering Intern

Mar 2023 - Jan 2025, Fargo, ND

- Rescued a sensor program at risk of cancellation: took a mechanical design from non-functional to working and built a heat map algorithm that improved accuracy from roughly $\pm 60\text{-}80\%$ to $\pm 5\%$
- Removed a calibration step that skewed the reference value and corrected for product variation across materials
- Owned lab and field testing, test stands, and data analysis for the project, writing tools in Python and C#

Faul Building and Repair: Foreman

Summer 2021, Fargo, ND

- Led a crew of up to 8 on demolition, framing, and repair jobs; handled scheduling and served as the crew's contact with the owner and customers

Co-founder, small business

2024 - Present, Fargo, ND

- Building low-cost, easy-to-use technical solutions for farmers

EDUCATION

North Dakota State University: B.S. Mechanical Engineering

December 2024